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Paper 1: Predictive AI in Clinical Pharmacology: A Call to Action to Develop Robust Bench

Bibliographic Information

- **Title:** Predictive AI in Clinical Pharmacology: A Call to Action to Develop Robust Benchmarking Practices
- **Authors:** Jane Knöchel, Sven Mensing, et al.
- **Journal:** Clinical Pharmacology & Therapeutics, 2025
- **DOI:** N/A

Paper Type: GENERIC

Quick Take

This perspective paper argues that the lack of standardized benchmarking datasets and evaluation protocols is a critical barrier to adopting AI/ML in clinical pharmacology. The authors propose creating a public repository of realistic benchmarking datasets, modeled after ImageNet, to enable fair comparison of methods and build trust. A working group has been formed to drive this initiative.

Executive Summary

This perspective paper argues that the lack of standardized benchmarking datasets and evaluation protocols is a critical barrier to adopting AI/ML methods in clinical pharmacology. The authors identify four key challenges: inconsistent metrics, unrealistic synthetic data, poor reproducibility, and divergent evaluation paradigms between ML and NLME approaches. They propose creating a public repository of realistic benchmarking datasets (both real and synthetic) with standardized evaluation tasks, modeled after successful initiatives like ImageNet. A working group has been formed to drive this effort, and the paper calls for interdisciplinary collaboration to build infrastructure that enables fair comparison, builds trust, and accelerates regulatory acceptance of AI-driven predictive models.

Scientific Context & Motivation

AI/ML methods are increasingly proposed for predictive modeling in clinical pharmacology, but their adoption is hindered by the lack of standardized evaluation frameworks. Unlike fields like computer vision (ImageNet) or medical imaging (MedSegBench), clinical pharmacology lacks public benchmarking datasets that reflect real-world data complexities (missingness, irregular sampling, outliers). This leads to fragmented evaluation practices, over-optimistic performance claims on synthetic data, and difficulty comparing methods across studies. Regulatory bodies require adequate validation, but without common benchmarks, it is challenging to establish

confidence in new AI approaches. This paper addresses this gap by calling for a community-driven effort to build benchmarking infrastructure.

✂ Methodological Snapshot

This is a perspective paper, not a methods paper. It does not present new methodology but rather synthesizes existing challenges and proposes a framework for standardized benchmarking. The authors draw on their collective experience as practitioners implementing AI in clinical pharmacology. They identify four challenges through informal observation of the literature and their own work. The proposed solution—a public repository of benchmarking datasets with standardized evaluation tasks—is described at a high level, with references to successful analogs in other fields (ImageNet, MedSegBench). A working group has been formed to operationalize this vision.

📋 Detailed Analysis

The authors identify four fundamental challenges that collectively undermine progress in applying AI to clinical pharmacology: (1) inconsistent performance metrics across studies, (2) reliance on synthetic or simplified data that fails to capture real-world complexities (missingness, irregular sampling, outliers), (3) poor reproducibility due to unavailable or non-functional code, and (4) divergent evaluation protocols between ML (out-of-sample, nested cross-validation) and NLME (in-sample diagnostics like VPC, NPDE). The proposal to create a public repository of benchmarking datasets—both real and carefully crafted synthetic—mirrors successful initiatives like ImageNet in computer vision. The authors argue that standardized benchmarks would reduce uncertainty, enabling analysts to confidently explore advanced methods without compromising patient safety. The paper also notes that regulatory bodies require adequate validation, and standardized benchmarks would facilitate regulatory acceptance. A working group has been formed to advance this initiative, with a call for interdisciplinary collaboration.

Domain Context

This perspective sits at the intersection of pharmacometrics, machine learning, and regulatory science. It addresses a critical gap: while AI/ML methods proliferate in clinical pharmacology, there is no equivalent of the FDA’s ‘Model-Informed Drug Development’ framework for evaluating these new approaches. The paper bridges the traditional NLME community (focused on mechanistic models) and the data-driven ML community, advocating for a common evaluation language. The reference to ImageNet is particularly apt—that dataset catalyzed a revolution in computer vision by providing a common benchmark. The authors extend a recent proposal by Sale and Liang (2024) for annual PK/PD benchmarking, broadening the scope to a permanent repository usable during method development.

🔑 Key Findings

The paper identifies four fundamental challenges hindering AI adoption in clinical pharmacology: (1) inconsistent performance metrics across studies, (2) reliance on synthetic data that fails to

capture real-world complexity, (3) poor reproducibility due to unavailable or non-functional code, and (4) divergent evaluation protocols between ML (out-of-sample) and NLME (in-sample) frameworks. The authors propose a public repository of benchmarking datasets with standardized evaluation tasks, modeled after ImageNet. They argue that such infrastructure would reduce uncertainty, enable fair comparisons, build trust, and accelerate regulatory acceptance. A working group has been formed to advance this initiative.

Strengths & Limitations

Strengths

- Timely and relevant: addresses a critical gap that is widely recognized but not yet systematically addressed.
- Clear identification of four fundamental challenges that hinder AI adoption.
- Proposes a concrete, actionable solution (public benchmarking repository) with successful analogs in other fields.
- Emphasizes the importance of realistic synthetic data that captures real-world complexities.
- Calls for interdisciplinary collaboration and has already formed a working group to move forward.
- Balanced perspective: acknowledges the need for conservatism in patient-facing applications while advocating for reducing unnecessary uncertainty.

Limitations (Acknowledged by Authors)

- Data privacy concerns necessitate the use of synthetic data, which must be carefully crafted to mirror real-world complexities.
- The initiative requires significant interdisciplinary collaboration and long-term commitment from stakeholders.
- The authors acknowledge that building a repository akin to ImageNet is a major undertaking.

Limitations (Expert Review)

- The paper does not provide concrete details on how the benchmarking datasets will be curated, validated, or maintained over time.
- The proposal lacks a discussion of potential biases in benchmark design (e.g., which tasks are prioritized, which populations are represented).
- The paper does not address how to handle proprietary data that cannot be shared, which may limit the real-data component of the repository.
- The call for out-of-sample evaluation for all frameworks, while methodologically sound, may be challenging for NLME models where in-sample diagnostics are deeply ingrained in practice.

Generalizability

The proposed framework is intended to be generalizable across clinical pharmacology applications, including population PK, PKPD, and potentially systems pharmacology. However, the specific implementation will depend on the tasks and datasets chosen by the working group.

Figures & Tables

- **Figure 1:** Highlights key differences between synthetic and real clinical data, including missingness, irregular sampling/dosing, and outliers.
 - *Significance:* Illustrates the core problem: synthetic data often glosses over real-world complexities, leading to overly optimistic model performance assessments.
 - **Table 1:** Summarizes strategic impacts of the proposed benchmarking initiative across future trends, long-term vision, interdisciplinary impact, and regulatory support.
 - *Significance:* Provides a concise overview of the expected benefits, helping stakeholders understand the value proposition.
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Code & Reproducibility Assessment

The paper does not provide code or data, but it explicitly calls for public release of model code and evaluation pipelines as part of the proposed benchmarking initiative.

Future Directions

The immediate next step is for the newly formed working group to define a set of initial benchmarking tasks (e.g., population PK prediction, covariate effect estimation) and release a first version of the repository with both real and synthetic datasets. Subsequent efforts should include: (1) developing guidelines for creating realistic synthetic data that captures missingness, irregular sampling, and outliers; (2) establishing a community process for adding new tasks and datasets; (3) creating a leaderboard for model comparisons; (4) engaging regulatory agencies to align on acceptable validation standards; and (5) integrating the benchmarking framework into educational curricula to train the next generation of pharmacometricians.

Expert Commentary

This is a timely and necessary call to action. The fragmentation in evaluation practices is a real bottleneck—I’ve seen promising ML methods fail to gain traction simply because their performance claims couldn’t be independently verified. The authors correctly identify that the gap between synthetic and real data is a major source of over-optimism. However, building a benchmarking repository is a massive undertaking: curating realistic synthetic data that captures the messiness of clinical data (missingness, irregular sampling, outliers) is non-trivial. The working group will need to address governance, data privacy, and long-term maintenance. The comparison to ImageNet is aspirational but apt—that dataset required years of coordinated effort. I’d add that the initiative should also consider benchmarking computational efficiency, as clinical pharmacology often requires rapid turnaround. The authors’ emphasis on out-of-sample evaluation for all frameworks is a key methodological point that should become standard practice.

Bottom Line

The field urgently needs a community-driven repository of realistic, public benchmarking datasets and standardized evaluation protocols to enable fair comparison of AI, ML, NLME,

and hybrid models in clinical pharmacology. Without such infrastructure, innovation is stifled, regulatory acceptance is hindered, and practitioners default to conservative methods, slowing progress toward precision medicine.